

Saige Howard

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Skills:

Laboratory Management; Supervising lab technicians; Training lab technicians; Maintaining inventory; Ordering supplies; Receiving and shipping packages; Organization of equipment and samples; Organization of databases; Compiling scientific reports; Operating analytical instruments; Analyzing and interpreting data for quaternary geochronology.

OSL sample collection, preparation, measurement, and analyzation; ESR sample collection, preparation, measurement, and analyzation; Organic material $\delta^{18}\text{O}$ and $\delta^2\text{H}$ sample preparation; Hydrologic Column Experiments; Batch Experiments; Water sampling; Macro charcoal preparation and counting; Sediment core extraction; Grain size analysis; Optical Mineralogy; Operation of heavy machinery; rock drill, rock saw, vibration auger; Quaternary geologic mapping; Soil descriptions.

Proficient in Microsoft Suite; Basic Python; Adobe Illustrator.

Laboratory training; Chemical safety training; Hazardous chemical training; Operation of analysis instruments; Operation of lab equipment.

Work Experience:

Luminescence Laboratory, Desert Research Institute, Reno, NV

Sample Preparation Laboratory Manager

2021 - Present

- Organizing and maintaining the laboratory space; inventory, ordering, and upholding chemical hygiene standards.
- Supervising lab technicians, delegating tasks, and managing schedules.
- Training new technicians in sediment sample preparation and laboratory housekeeping.
- Intake, tracking, and archiving samples received for OSL dating in the laboratory and the office database.
- Processing sediment samples in low light conditions using hazardous chemicals such as high-concentration Hydrofluoric acid and Hydrochloric acid.
- Communicating with clients using services offered by our lab about sample progress and characteristics.
- Operating RISØ and LexyG analysis instruments, rock core drills, precision rock core saw, rock saw, and centrifuge.

Nevada Robotics, Desert Research Institute, Reno, NV

Office Coordinator

2024 - 2025

- Organizing and maintaining the office and storage spaces for teaching materials, books, and robots used for the Nevada Robotics STEM Education Program.
- Supervising undergraduate Robotics Outreach Crew members, delegating tasks, managing schedules, and receiving inventory orders.

- Training new Robotics Outreach Crew members in office operations, preparing teaching materials and setting-up robots; and community outreach events, teacher trainings and tabling.
- Maintaining the robot lending library used by the Nevada Robotics program, where K-12 teachers can check out robots to use in the classroom.

United States Geological Survey, NAGT Summer Fellowship Program, Reston, VA

Intern Geologist

Fall 2023

- Prepared and analyzed samples for macro charcoal counting and grain size.
- Compiled reports on grain size and macro charcoal for multiple sediment cores extracted from the Potomac River used in flood reconstruction projects.
- Organized, cleaned, and set up a new laboratory space designated for macro charcoal and grain size analysis.
- Compiled reports of historical floods from newspaper databases, these reports were used to begin a mapping project of historical floods in GIS.

Subsurface Biogeochemical Laboratory, Desert Research Institute, Reno, NV

Laboratory Technician

2022 - 2023

- Performed hydrologic column and batch experiments, sampling for Arsenic and metal-organic frameworks.
- Produced reports on adsorption isotherms and material characterization curves of Iron Metal-organic Framework.
- Coordinated with external labs to analyze samples for Arsenic concentration.

Field Experience:

Brachiopod Sampling, Dr. Paula Noble

Collected brachiopods for $\delta^{18}\text{O}$ analysis for paleotemperature proxy in the Gabbs Mountains located in central Nevada.

Water Sampling for Stable Isotope Analysis, Dr. Scott Allen

Collected water samples to test for $\delta^{18}\text{O}$ and $\delta^2\text{H}$.

Ichthyosaur Excavation and Sampling for Stable Isotopes, Dr. Paula Noble

Collected samples to create an organic $\delta^{13}\text{C}$ profile for a location in Mina, Nevada.

Field Camp, Dr. Andrew Zuza

Created geologic maps for three different areas over the course of three weeks for my capstone course.

Sediment Core Extraction, Dr. Michael Toomey

Collected cores using a auger in Black Pond, VA, used for paleo flood reconstruction.

Basalt Bombs, Dr. Kathleen Rodrigues

Collected basalt bombs in Soda Lakes, NV to test for Thermoluminescence signals.

Publications and Contributions:

Saige Howard, Gary McGaughey, Paula Noble, and Randy Irmis, 2021 "Ichthy Isotopes"
Undergraduate Thesis Poster. Presented at the 2021 Fall Wolf Pack Discoveries Event
Porraket Dechdacho, Saige Howard, Ronald L. Hershey, Rishi Parashar, and Lazaro J. Perez,
2023, "Effective removal of arsenic from contaminated groundwater using an iron-based
metal organic framework" <https://doi.org/10.1016/j.eti.2023.103406>
Gary McGaughey, Randall B. Irmis, Neil P. Kelley, Saige Howard, Tylor Birthisel, Kathleen
Ritterbush, Lydia Tackett, Montana Hodges, and Paula J. Noble, submitted 2025, "Giant
ichthyosaurs survived until the end-Triassic mass extinction"

Education:

University of Nevada, Reno, NV

Fall 2025 - Present

Masters of Science in Geology

Thesis: Refining the Glacial Chronology of the Sierra Nevada Mountain Range Using
Luminescence and Electron Spin Resonance Dating Techniques

Advisor: Dr. Kathleen Rodrigues

University of Nevada, Reno, NV

Graduated 2022

Bachelor of Science in Geology

Senior Thesis: Organic δ^{13} Carbon profile of an Ichthyosaur quarry in Mina, NV
Employing the existing datasets of organic δ^{13} carbon isotopes of the field
location, the isotopic curve constructed from the quarry and tested it for organic
 δ^{13} carbon to place the fossil stratigraphically.

Advisor: Dr. Paula Noble

Robert McQueen High School, Reno, NV

Graduated 2018

Honors Diploma

Nevada Seal of Biliteracy, Spanish

Awards and Grants:

Millennium Scholarship, University of Nevada, Reno, 2018-2022

Daugherty-Westfall Undergraduate Research Award, University of Nevada, Reno, 2021

Nevada Undergraduate Research Award, University of Nevada, Reno, Fall 2022

Nomination for USGS NAGT Summer Internship, University of Nevada, Reno Summer 2022

AGU Travel Grant, New Orleans, LA 2025